

cv_validated_ensemble

scalars_validated_ensemble_cv

key	value
skipped_empty_gt	0
main_kwargs_applied.substitute	False
main_kwargs_applied.normalize_final_letters	True
step2_stats.n_models	9
step2_stats.spearman	1
step2_stats.kendall_b	1
step2_stats.spearman_exact_p_onetailed	2.7557e-06
step2_stats.kendall_exact_p_onetailed	2.7557e-06
step2_stats.p_method	exact
step2_stats.n_permutations	362880
step2_stats.min_attainable_p	2.7557e-06
step2_stats.pearson	0.9953
step2_stats.pearson_p	2.3385e-08
step2_stats.mean_wer_compression	0.8931
step3_determinism.identical_outputs	10495
step3_determinism.n	10495
step3_determinism.deterministic	True
step3_determinism.main_kwargs.substitute	False
step3_determinism.main_kwargs.normalize_final_letters	True
step4_resolution_true.reference	true
step4_resolution_true.n_boot	10000
step4_resolution_true.n_samples	10500
step4_resolution_true.min_resolved_gap_wer	0.6214
step4_resolution_true.max_unresolved_gap_wer	0
step4_resolution_true.n_resolved	36
step4_resolution_true.n_pairs	36
step4_resolution_pseudo.reference	pseudo
step4_resolution_pseudo.n_boot	10000
step4_resolution_pseudo.n_samples	10495
step4_resolution_pseudo.min_resolved_gap_wer	0.6839
step4_resolution_pseudo.max_unresolved_gap_wer	0
step4_resolution_pseudo.n_resolved	36
step4_resolution_pseudo.n_pairs	36
step8_permutation_summary.canonical_order	seamless_m4t+whisper_v3+fastconformer+w2vbert+hubert+wav2vec_sb+cohere
step8_permutation_summary.n_orders	6

key	value
step8_permutation_summary.n_transcripts	10495
step8_permutation_summary.total_differing	0
step8_permutation_summary.invariant	True
step8_permutation_summary.main_kwargs.substitute	False
step8_permutation_summary.main_kwargs.normalize_final_letters	True

step0_cross_wer_validated_ensemble_cv

	seamless_m4t	whisper_v3	whisper_v2	fastconformer	xlsr	w2vbert	hubert	wav2vec_sb	cohere
seamless_m4t	0	15.8	19.4	11.85	24.88	25.77	13.86	30.72	10.23
whisper_v3	15.8	0	14.77	15	25.5	26.23	17.06	31.71	13.88
whisper_v2	19.4	14.77	0	18.63	28.03	28.83	20.5	33.99	17.57
fastconformer	11.85	15	18.63	0	22.77	24.06	10.3	28.8	7.05
xlsr	24.88	25.5	28.03	22.77	0	31.68	23.49	31.1	22.29
w2vbert	25.77	26.23	28.83	24.06	31.68	0	25.33	37.2	23.32
hubert	13.86	17.06	20.5	10.3	23.49	25.33	0	29.58	9.23
wav2vec_sb	30.72	31.71	33.99	28.8	31.1	37.2	29.58	0	28.81
cohere	10.23	13.88	17.57	7.05	22.29	23.32	9.23	28.81	0

step0_err_corr_validated_ensemble_cv

	seamless_m4t	whisper_v3	whisper_v2	fastconformer	xlsr	w2vbert	hubert	wav2vec_sb	cohere
seamless_m4t	1	0.493	0.4572	0.4468	0.3933	0.3559	0.3527	0.3807	0.4864
whisper_v3	0.493	1	0.6827	0.4367	0.4573	0.439	0.3487	0.4105	0.4442
whisper_v2	0.4572	0.6827	1	0.3904	0.4453	0.4221	0.3313	0.4116	0.4045
fastconformer	0.4468	0.4367	0.3904	1	0.41	0.3553	0.4155	0.4007	0.5511
xlsr	0.3933	0.4573	0.4453	0.41	1	0.4547	0.4217	0.6335	0.3863
w2vbert	0.3559	0.439	0.4221	0.3553	0.4547	1	0.3421	0.4296	0.3379
hubert	0.3527	0.3487	0.3313	0.4155	0.4217	0.3421	1	0.4044	0.3997
wav2vec_sb	0.3807	0.4105	0.4116	0.4007	0.6335	0.4296	0.4044	1	0.3594
cohere	0.4864	0.4442	0.4045	0.5511	0.3863	0.3379	0.3997	0.3594	1

step0_flags_validated_ensemble_cv

pair	cross_wer	err_corr	note
seamless_m4t fastconformer	11.85	0.4468	unusually similar hypotheses - suspect shared lab/data/teacher
seamless_m4t hubert	13.86	0.3527	unusually similar hypotheses - suspect shared lab/data/teacher
seamless_m4t cohere	10.23	0.4864	unusually similar hypotheses - suspect shared lab/data/teacher

pair	cross_wer	err_corr	note
whisper_v3 cohere	13.88	0.4442	unusually similar hypotheses - suspect shared lab/data/teacher
fastconformer hubert	10.3	0.4155	unusually similar hypotheses - suspect shared lab/data/teacher
fastconformer cohere	7.05	0.5511	unusually similar hypotheses - suspect shared lab/data/teacher
hubert cohere	9.23	0.3997	unusually similar hypotheses - suspect shared lab/data/teacher

step1_bias_validated_ensemble_cv

model	role	n_samples	spearman_full	spearman_loo	kendall_full	kendall_loo	delta_spearman	delta_kendall
cohere	voter	10495	0.5883	0.5824	0.5633	0.5539	-0.0059	-0.0094
fastconformer	voter	10495	0.7077	0.7088	0.6771	0.6755	0.0011	-0.0016
hubert	voter	10495	0.721	0.7077	0.6907	0.674	-0.0133	-0.0167
seamless_m4t	voter	10495	0.7979	0.8062	0.763	0.7692	0.0084	0.0061
whisper_v3	voter	10495	0.8672	0.8824	0.8275	0.8449	0.0152	0.0174
whisper_v2	ranked_only(control)	10495	0.9034	0.9034	0.8641	0.8641	0	0
w2vbert	voter	10495	0.9228	0.9302	0.88	0.8892	0.0074	0.0091
xlsr	ranked_only(control)	10495	0.9279	0.9279	0.8876	0.8876	0	0
wav2vec_sb	voter	10495	0.9519	0.956	0.9144	0.9205	0.0041	0.0061

step2_ranking_validated_ensemble_cv

model	role	n_samples	wer_true	cer_true	wer_pseudo_loo	cer_pseudo_loo	rank_true	rank_pseudo	wer_compression
cohere	voter	10496	5.5591	1.5028	5.4842	1.6077	1	1	0.0749
fastconformer	voter	10496	7.1898	2.158	6.8721	2.0654	2	2	0.3178
hubert	voter	10495	7.8081	2.1291	9.0306	2.411	3	3	-1.2225
seamless_m4t	voter	10495	10.7671	3.4897	9.7145	3.3357	4	4	1.0526
whisper_v3	voter	10495	14.7106	4.636	13.0371	4.1872	5	5	1.6735
whisper_v2	ranked_only	10495	18.4155	6.0072	16.4124	5.4924	6	6	2.003
xlsr	ranked_only	10495	22.8174	6.433	21.3202	5.9605	7	7	1.4972
w2vbert	voter	10495	23.9822	6.5829	22.6039	6.2056	8	8	1.3783
wav2vec_sb	voter	10495	29.1187	9.6559	27.8553	9.2316	9	9	1.2634

step3_model_dropout_validated_ensemble_cv

model	times_filtered_out	times_in_pool	filtered_out_rate
wav2vec_sb	38946	73467	0.5301
w2vbert	29796	73467	0.4056
whisper_v3	13227	73467	0.18
seamless_m4t	8356	73467	0.1137

model	times_filtered_out	times_in_pool	filtered_out_rate
hubert	6225	73467	0.0847
fastconformer	4157	73466	0.0566
cohere	2447	73466	0.0333

step3_runs_validated_ensemble_cv

run	n_voters	n_samples	filter_fire_rate	tie_rate	refsel_longest_match_validation	refsel_most_common_words
full_pool	7	10495	0.8568	0.0406	0.7316	0.2684
LOO_minus_seamless_m4t	6	10495	0.8332	0.0344	0.7644	0.2356
LOO_minus_whisper_v3	6	10495	0.8289	0.0309	0.7683	0.2317
LOO_minus_fastconformer	6	10496	0.838	0.0369	0.757	0.243
LOO_minus_w2vbert	6	10495	0.7719	0.0263	0.8016	0.1984
LOO_minus_hubert	6	10495	0.8388	0.0353	0.7597	0.2403
LOO_minus_wav2vec_sb	6	10495	0.7447	0.0258	0.8188	0.1812
LOO_minus_cohere	6	10496	0.8396	0.0407	0.7487	0.2513

step4_pairs_pseudo_validated_ensemble_cv

pair	adjacent	gap_wer	preserved_frac	resolved_95
hubert < seamless_m4t	True	0.6839	0.9996	True
xlsr < w2vbert	True	1.2837	1	True
cohere < fastconformer	True	1.3879	1	True
fastconformer < hubert	True	2.1603	1	True
fastconformer < seamless_m4t	False	2.8442	1	True
seamless_m4t < whisper_v3	True	3.3226	1	True
whisper_v3 < whisper_v2	True	3.3753	1	True
cohere < hubert	False	3.5482	1	True
hubert < whisper_v3	False	4.0065	1	True
cohere < seamless_m4t	False	4.2321	1	True
whisper_v2 < xlsr	True	4.9077	1	True
w2vbert < wav2vec_sb	True	5.2514	1	True
fastconformer < whisper_v3	False	6.1668	1	True
whisper_v2 < w2vbert	False	6.1915	1	True
xlsr < wav2vec_sb	False	6.5351	1	True
seamless_m4t < whisper_v2	False	6.6979	1	True
hubert < whisper_v2	False	7.3818	1	True
cohere < whisper_v3	False	7.5547	1	True
whisper_v3 < xlsr	False	8.2831	1	True
fastconformer < whisper_v2	False	9.5421	1	True

pair	adjacent	gap_wer	preserved_frac	resolved_95
whisper_v3 < w2vbert	False	9.5668	1	True
cohere < whisper_v2	False	10.93	1	True
whisper_v2 < wav2vec_sb	False	11.4428	1	True
seamless_m4t < xlsr	False	11.6056	1	True
hubert < xlsr	False	12.2895	1	True
seamless_m4t < w2vbert	False	12.8894	1	True
hubert < w2vbert	False	13.5733	1	True
fastconformer < xlsr	False	14.4498	1	True
whisper_v3 < wav2vec_sb	False	14.8182	1	True
fastconformer < w2vbert	False	15.7336	1	True
cohere < xlsr	False	15.8378	1	True
cohere < w2vbert	False	17.1215	1	True
seamless_m4t < wav2vec_sb	False	18.1407	1	True
hubert < wav2vec_sb	False	18.8247	1	True
fastconformer < wav2vec_sb	False	20.985	1	True
cohere < wav2vec_sb	False	22.3729	1	True

step4_pairs_true_validated_ensemble_cv

pair	adjacent	gap_wer	preserved_frac	resolved_95
fastconformer < hubert	True	0.6214	1	True
xlsr < w2vbert	True	1.1621	1	True
cohere < fastconformer	True	1.6296	1	True
cohere < hubert	False	2.2511	1	True
hubert < seamless_m4t	True	2.9664	1	True
fastconformer < seamless_m4t	False	3.5878	1	True
whisper_v3 < whisper_v2	True	3.7042	1	True
seamless_m4t < whisper_v3	True	3.9295	1	True
whisper_v2 < xlsr	True	4.3988	1	True
w2vbert < wav2vec_sb	True	5.1348	1	True
cohere < seamless_m4t	False	5.2174	1	True
whisper_v2 < w2vbert	False	5.561	1	True
xlsr < wav2vec_sb	False	6.2969	1	True
hubert < whisper_v3	False	6.8958	1	True
fastconformer < whisper_v3	False	7.5173	1	True
seamless_m4t < whisper_v2	False	7.6337	1	True
whisper_v3 < xlsr	False	8.103	1	True
cohere < whisper_v3	False	9.1469	1	True
whisper_v3 < w2vbert	False	9.2652	1	True

pair	adjacent	gap_wer	preserved_frac	resolved_95
hubert < whisper_v2	False	10.6	1	True
whisper_v2 < wav2vec_sb	False	10.6958	1	True
fastconformer < whisper_v2	False	11.2215	1	True
seamless_m4t < xlsr	False	12.0325	1	True
cohere < whisper_v2	False	12.8511	1	True
seamless_m4t < w2vbert	False	13.1947	1	True
whisper_v3 < wav2vec_sb	False	14.4	1	True
hubert < xlsr	False	14.9989	1	True
fastconformer < xlsr	False	15.6203	1	True
hubert < w2vbert	False	16.161	1	True
fastconformer < w2vbert	False	16.7824	1	True
cohere < xlsr	False	17.2499	1	True
seamless_m4t < wav2vec_sb	False	18.3295	1	True
cohere < w2vbert	False	18.4121	1	True
hubert < wav2vec_sb	False	21.2958	1	True
fastconformer < wav2vec_sb	False	21.9172	1	True
cohere < wav2vec_sb	False	23.5469	1	True

step5_baselines_validated_ensemble_cv

reference	n_models	spearman	kendall_b	spearman_exact_p_onetailed	kendall_exact_p_onetailed	p_method	n_permutations	min_attainable_p	mean_abs_wer_err
LOO ensemble (proposed)	9	1	1	2.7557e-06	2.7557e-06	exact	362880	2.7557e-06	1.1648
full ensemble (biased)	9	1	1	2.7557e-06	2.7557e-06	exact	362880	2.7557e-06	1.8296
single model: seamless_m4t	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	2.8048
LOO ensemble matched to n-1 (vs seamless_m4t)	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	1.1788
single model: whisper_v3	8	0.8333	0.7143	0.0077	0.0071	exact	40320	2.4802e-05	5.1675
LOO ensemble matched to n-1 (vs whisper_v3)	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	1.1012
single model: fastconformer	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	0.7331
LOO ensemble matched to n-1 (vs fastconformer)	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	1.2707
single model: w2vbert	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	13.2048
LOO ensemble matched to n-1 (vs w2vbert)	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	1.1381
single model: hubert	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	2.003
LOO ensemble matched to n-1 (vs hubert)	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	1.1576
single model: wav2vec_sb	8	0.9286	0.8571	0.0011	0.0009	exact	40320	2.4802e-05	17.8119
LOO ensemble matched to n-1 (vs wav2vec_sb)	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	1.1525
single model: cohere	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	0.7447
LOO ensemble matched to n-1 (vs cohere)	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	1.301

step6_contribution_validated_ensemble_cv

model_removed	delta_ensemble_wer	delta_ranking_spearman	helps_reference_quality	helps_ranking	delta_pairs_wrong	delta_pairs_wrong_resolvable
seamless_m4t	0.2348	0	True	True	0	0
whisper_v3	-0.1804	0	False	True	0	0
fastconformer	0.5015	0	True	True	0	0
w2vbert	-0.0902	0	False	True	0	0
hubert	0.697	0	True	True	0	0
wav2vec_sb	-0.1184	0	False	True	0	0
cohere	0.8942	0	True	True	0	0

step6_pools_validated_ensemble_cv

pool	k	ensemble_wer_vs_true	ensemble_cer_vs_true	ranking_spearman	ranking_exact_p	ranking_kendall_b	ranking_kendall_exact_p	n_pairs_wrong	n_pairs_wrong_resolvable	worst_wrong_gap_wer	worst_wrong_pair
seamless_m4t+fastconformer+hubert+cohere	4	4.4357	1.2548	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+fastconformer+w2vbert+hubert+cohere	5	4.4752	1.2711	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+fastconformer+hubert+wav2vec_sb+cohere	5	4.5297	1.2993	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+fastconformer+hubert+cohere	5	4.586	1.2753	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+fastconformer+w2vbert+hubert+wav2vec_sb+cohere	6	4.6367	1.3214	1	2.7557e-06	1	2.7557e-06	0	0	0	
fastconformer+hubert+cohere	3	4.6443	1.2966								
seamless_m4t+whisper_v3+fastconformer+w2vbert+hubert+cohere	6	4.6987	1.2932	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+fastconformer+hubert+wav2vec_sb+cohere	6	4.7269	1.3221	1	2.7557e-06	1	2.7557e-06	0	0	0	
fastconformer+w2vbert+hubert+cohere	4	4.7927	1.3625	1	2.7557e-06	1	2.7557e-06	0	0	0	
whisper_v3+fastconformer+hubert+cohere	4	4.8133	1.3937	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+fastconformer+w2vbert+hubert+wav2vec_sb+cohere	7	4.8171	1.3666	1	2.7557e-06	1	2.7557e-06	0	0	0	
fastconformer+hubert+wav2vec_sb+cohere	4	4.849	1.394	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+hubert+cohere	3	4.8813	1.3839								
whisper_v3+fastconformer+w2vbert+hubert+cohere	5	4.8923	1.4309	1	2.7557e-06	1	2.7557e-06	0	0	0	
whisper_v3+fastconformer+hubert+wav2vec_sb+cohere	5	4.9204	1.4553	1	2.7557e-06	1	2.7557e-06	0	0	0	
fastconformer+w2vbert+hubert+wav2vec_sb+cohere	5	4.9505	1.4405	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+hubert+wav2vec_sb+cohere	4	5.0293	1.4153	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	1.6307	cohere fastconformer
whisper_v3+fastconformer+w2vbert+hubert+wav2vec_sb+cohere	6	5.0519	1.4732	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+w2vbert+hubert+cohere	4	5.0575	1.3932	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	1.6307	cohere fastconformer
seamless_m4t+fastconformer+hubert	3	5.1477	1.4789								
seamless_m4t+whisper_v3+w2vbert+hubert+cohere	5	5.1928	1.453	1	2.7557e-06	1	2.7557e-06	0	0	0	

pool	k	ensemble_wer_vs_true	ensemble_cer_vs_true	ranking_spearman	ranking_exact_p	ranking_kendall_b	ranking_kendall_exact_p	n_pairs_wrong	n_pairs_wrong_resolvable	worst_wrong_gap_wer	worst_wrong_pair
seamless_m4t+fastconformer+cohere	3	5.1969	1.4699								
seamless_m4t+whisper_v3+hubert+cohere	4	5.2099	1.4843	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	2.9664	hubert seamless_m4t
seamless_m4t+w2vbert+hubert+wav2vec_sb+cohere	5	5.2454	1.4686	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	1.6307	cohere fastconformer
seamless_m4t+fastconformer+wav2vec_sb+cohere	4	5.2473	1.464	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+hubert+wav2vec_sb+cohere	5	5.3074	1.4868	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+w2vbert+hubert+wav2vec_sb+cohere	6	5.3186	1.48	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+fastconformer+w2vbert+cohere	4	5.3244	1.4648	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+fastconformer+w2vbert+wav2vec_sb+cohere	5	5.4145	1.5398	1	2.7557e-06	1	2.7557e-06	0	0	0	
whisper_v3+hubert+cohere	3	5.4502	1.5759								
seamless_m4t+fastconformer+w2vbert+hubert	4	5.4539	1.547	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+fastconformer+hubert+wav2vec_sb	4	5.4915	1.6014	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+fastconformer+hubert	4	5.499	1.609	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+fastconformer+w2vbert+wav2vec_sb+cohere	6	5.5141	1.5519	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb+cohere	5	5.5216	1.5637	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+fastconformer+w2vbert+cohere	5	5.5291	1.547	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+fastconformer+cohere	4	5.5329	1.5836	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+fastconformer+hubert+wav2vec_sb	5	5.5779	1.6569	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+fastconformer+w2vbert+hubert	5	5.5835	1.6269	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+fastconformer+w2vbert+hubert+wav2vec_sb	5	5.5967	1.647	1	2.7557e-06	1	2.7557e-06	0	0	0	
whisper_v3+hubert+wav2vec_sb+cohere	4	5.6587	1.6067	0.9667	8.2672e-05	0.8889	0.0001	2	2	2.8948	hubert seamless_m4t
whisper_v3+w2vbert+hubert+cohere	4	5.685	1.5785	0.9667	8.2672e-05	0.8889	0.0001	2	2	2.8948	hubert seamless_m4t
seamless_m4t+whisper_v3+fastconformer+w2vbert+hubert+wav2vec_sb	6	5.7113	1.666	1	2.7557e-06	1	2.7557e-06	0	0	0	
whisper_v3+fastconformer+cohere	3	5.7413	1.6497								
whisper_v3+fastconformer+wav2vec_sb+cohere	4	5.8523	1.6687	1	2.7557e-06	1	2.7557e-06	0	0	0	
fastconformer+w2vbert+cohere	3	5.8598	1.6878								
whisper_v3+fastconformer+hubert	3	5.8785	1.7284								
w2vbert+hubert+cohere	3	5.886	1.7193								
whisper_v3+w2vbert+hubert+wav2vec_sb+cohere	5	5.9424	1.7052	0.9667	8.2672e-05	0.8889	0.0001	2	2	2.9571	hubert seamless_m4t
whisper_v3+fastconformer+w2vbert+cohere	4	5.9519	1.6676	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	0.5555	fastconformer hubert
fastconformer+w2vbert+wav2vec_sb+cohere	4	6.0045	1.7692	0.8667	0.0023	0.7778	0.0012	4	4	5.208	cohere seamless_m4t
whisper_v3+fastconformer+w2vbert+wav2vec_sb+cohere	5	6.1116	1.7665	1	2.7557e-06	1	2.7557e-06	0	0	0	
fastconformer+wav2vec_sb+cohere	3	6.1247	1.8354								

pool	k	ensemble_wer_vs_true	ensemble_cer_vs_true	ranking_spearman	ranking_exact_p	ranking_kendall_b	ranking_kendall_exact_p	n_pairs_wrong	n_pairs_wrong_resolvable	worst_wrong_gap_wer	worst_wrong_pair
w2vbert+hubert+wav2vec_sb+cohere	4	6.1284	1.8129	0.9167	0.0007	0.8333	0.0004	3	3	5.2097	cohere seamless_m4t
whisper_v3+fastconformer+hubert+wav2vec_sb	4	6.1547	1.8319	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	2.9571	hubert seamless_m4t
whisper_v3+fastconformer+w2vbert+hubert	4	6.1791	1.8175	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	2.9571	hubert seamless_m4t
whisper_v3+fastconformer+w2vbert+hubert+wav2vec_sb	5	6.3951	1.9274	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	2.9571	hubert seamless_m4t
hubert+wav2vec_sb+cohere	3	6.4083	2.0195								
fastconformer+w2vbert+hubert	3	6.4421	1.932								
seamless_m4t+w2vbert+wav2vec_sb+cohere	4	6.6375	1.8738	0.9333	0.0004	0.8333	0.0004	3	3	3.9434	seamless_m4t whisper_v3
seamless_m4t+whisper_v3+w2vbert+wav2vec_sb+cohere	5	6.6563	1.8776	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	1.6307	cohere fastconformer
seamless_m4t+whisper_v3+w2vbert+cohere	4	6.6968	1.8645	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	1.6296	cohere fastconformer
seamless_m4t+w2vbert+cohere	3	6.7043	1.9295								
seamless_m4t+whisper_v3+wav2vec_sb+cohere	4	6.7062	1.9417	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	1.6296	cohere fastconformer
fastconformer+w2vbert+hubert+wav2vec_sb	4	6.722	2.056	0.95	0.0002	0.8889	0.0001	2	2	3.5789	fastconformer seamless_m4t
seamless_m4t+whisper_v3+cohere	3	6.7419	1.9664								
seamless_m4t+wav2vec_sb+cohere	3	6.8132	2.0637								
seamless_m4t+whisper_v3+w2vbert+hubert+wav2vec_sb	5	6.9681	2.0381	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+hubert	3	6.9953	2.0945								
seamless_m4t+whisper_v3+hubert+wav2vec_sb	4	7.0076	2.0728	1	2.7557e-06	1	2.7557e-06	0	0	0	
seamless_m4t+whisper_v3+w2vbert+hubert	4	7.0095	1.967	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	2.8948	hubert seamless_m4t
fastconformer+hubert+wav2vec_sb	3	7.0508	2.2523								
seamless_m4t+whisper_v3+fastconformer+w2vbert+wav2vec_sb	5	7.0959	2.0686	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	0.6219	fastconformer hubert
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb	4	7.1372	2.0526	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	0.5591	fastconformer hubert
seamless_m4t+whisper_v3+fastconformer	3	7.1455	2.1181								
seamless_m4t+whisper_v3+fastconformer+w2vbert	4	7.1748	2.021	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	0.5591	fastconformer hubert
seamless_m4t+fastconformer+w2vbert+wav2vec_sb	4	7.1955	2.1188	0.9667	8.2672e-05	0.8889	0.0001	2	2	3.9434	seamless_m4t whisper_v3
seamless_m4t+fastconformer+w2vbert	3	7.2236	2.1039								
seamless_m4t+w2vbert+hubert	3	7.357	2.1553								
seamless_m4t+w2vbert+hubert+wav2vec_sb	4	7.3852	2.2048	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	3.9434	seamless_m4t whisper_v3
seamless_m4t+fastconformer+wav2vec_sb	3	7.4566	2.2968								
seamless_m4t+hubert+wav2vec_sb	3	7.7816	2.4806								
whisper_v3+w2vbert+wav2vec_sb+cohere	4	8.1761	2.3783	0.9	0.001	0.8333	0.0004	3	3	5.2097	cohere seamless_m4t
whisper_v3+wav2vec_sb+cohere	3	8.5686	2.658								
whisper_v3+w2vbert+cohere	3	8.7113	2.5595								
whisper_v3+fastconformer+w2vbert+wav2vec_sb	4	8.7116	2.5963	0.95	0.0002	0.8889	0.0001	2	2	3.5789	fastconformer seamless_m4t
whisper_v3+w2vbert+hubert+wav2vec_sb	4	8.9295	2.6933	0.9667	8.2672e-05	0.8889	0.0001	2	2	3.7067	whisper_v3 whisper_v2
whisper_v3+fastconformer+wav2vec_sb	3	9.1493	2.8489								
whisper_v3+fastconformer+w2vbert	3	9.2433	2.715								
whisper_v3+w2vbert+hubert	3	9.3146	2.7523								

pool	k	ensemble_wer_vs_true	ensemble_cer_vs_true	ranking_spearman	ranking_exact_p	ranking_kendall_b	ranking_kendall_exact_p	n_pairs_wrong	n_pairs_wrong_resolvable	worst_wrong_gap_wer	worst_wrong_pair
whisper_v3+hubert+wav2vec_sb	3	9.4875	3.0239								
seamless_m4t+whisper_v3+w2vbert+wav2vec_sb	4	9.5457	2.788	0.9833	2.4802e-05	0.9444	2.4802e-05	1	1	3.7067	whisper_v3 whisper_v2
seamless_m4t+whisper_v3+w2vbert	3	10.1175	3.0218								
seamless_m4t+whisper_v3+wav2vec_sb	3	10.1532	3.1887								
w2vbert+wav2vec_sb+cohere	3	10.5715	3.345								
fastconformer+w2vbert+wav2vec_sb	3	11.1145	3.5478								
w2vbert+hubert+wav2vec_sb	3	11.3286	3.6741								
seamless_m4t+w2vbert+wav2vec_sb	3	12.1214	3.8221								
whisper_v3+w2vbert+wav2vec_sb	3	13.9983	4.427								

step6_wrong_pairs_validated_ensemble_cv

pool	k	pair	true_winner	pred_winner	wer_true_a	wer_true_b	wer_pseudo_a	wer_pseudo_b	gap_wer_true	gap_wer_pseudo	human_resolvable
whisper_v3+w2vbert+wav2vec_sb+cohere	4	cohere seamless_m4t	cohere	seamless_m4t	5.5591	10.7688	13.1172	11.1912	5.2097	1.9259	True
w2vbert+hubert+wav2vec_sb+cohere	4	cohere seamless_m4t	cohere	seamless_m4t	5.5591	10.7688	11.3123	10.9073	5.2097	0.405	True
fastconformer+w2vbert+wav2vec_sb+cohere	4	cohere seamless_m4t	cohere	seamless_m4t	5.5591	10.7671	10.616	10.4379	5.208	0.1781	True
seamless_m4t+fastconformer+w2vbert+wav2vec_sb	4	seamless_m4t whisper_v3	seamless_m4t	whisper_v3	10.7688	14.7122	14.3939	13.5712	3.9434	0.8226	True
seamless_m4t+w2vbert+hubert+wav2vec_sb	4	seamless_m4t whisper_v3	seamless_m4t	whisper_v3	10.7688	14.7122	14.9734	14.1458	3.9434	0.8276	True
seamless_m4t+w2vbert+wav2vec_sb+cohere	4	seamless_m4t whisper_v3	seamless_m4t	whisper_v3	10.7688	14.7122	13.794	13.2559	3.9434	0.5381	True
seamless_m4t+whisper_v3+w2vbert+wav2vec_sb	4	whisper_v3 whisper_v2	whisper_v3	whisper_v2	14.7122	18.4189	16.2696	16.1695	3.7067	0.1	True
whisper_v3+w2vbert+hubert+wav2vec_sb	4	whisper_v3 whisper_v2	whisper_v3	whisper_v2	14.7122	18.4189	16.976	16.9304	3.7067	0.0456	True
whisper_v3+fastconformer+w2vbert+wav2vec_sb	4	fastconformer seamless_m4t	fastconformer	seamless_m4t	7.1898	10.7688	13.9938	11.7111	3.5789	2.2827	True
fastconformer+w2vbert+hubert+wav2vec_sb	4	fastconformer seamless_m4t	fastconformer	seamless_m4t	7.1898	10.7688	12.119	11.4996	3.5789	0.6194	True
fastconformer+w2vbert+wav2vec_sb+cohere	4	fastconformer seamless_m4t	fastconformer	seamless_m4t	7.1898	10.7671	11.1109	10.4379	3.5773	0.673	True
seamless_m4t+whisper_v3+hubert+cohere	4	hubert seamless_m4t	hubert	seamless_m4t	7.874	10.8403	10.1524	10.0408	2.9664	0.1116	True
whisper_v3+fastconformer+w2vbert+hubert	4	hubert seamless_m4t	hubert	seamless_m4t	7.8117	10.7688	11.8474	10.5429	2.9571	1.3046	True
whisper_v3+fastconformer+hubert+wav2vec_sb	4	hubert seamless_m4t	hubert	seamless_m4t	7.8117	10.7688	11.5879	10.6456	2.9571	0.9423	True
whisper_v3+w2vbert+hubert+wav2vec_sb	4	hubert seamless_m4t	hubert	seamless_m4t	7.8117	10.7688	15.6075	12.2453	2.9571	3.3622	True
fastconformer+w2vbert+hubert+wav2vec_sb	4	hubert seamless_m4t	hubert	seamless_m4t	7.8117	10.7688	13.0963	11.4996	2.9571	1.5967	True
w2vbert+hubert+wav2vec_sb+cohere	4	hubert seamless_m4t	hubert	seamless_m4t	7.8117	10.7688	12.7377	10.9073	2.9571	1.8304	True
whisper_v3+fastconformer+w2vbert+hubert+wav2vec_sb	5	hubert seamless_m4t	hubert	seamless_m4t	7.8117	10.7688	11.2815	10.5903	2.9571	0.6912	True
whisper_v3+w2vbert+hubert+wav2vec_sb+cohere	5	hubert seamless_m4t	hubert	seamless_m4t	7.8117	10.7688	11.0274	10.1731	2.9571	0.8544	True
seamless_m4t+whisper_v3+w2vbert+hubert	4	hubert seamless_m4t	hubert	seamless_m4t	7.874	10.7688	12.7393	12.5226	2.8948	0.2167	True
whisper_v3+w2vbert+hubert+cohere	4	hubert seamless_m4t	hubert	seamless_m4t	7.874	10.7688	11.5506	10.0577	2.8948	1.4929	True
whisper_v3+hubert+wav2vec_sb+cohere	4	hubert seamless_m4t	hubert	seamless_m4t	7.874	10.7688	11.2876	10.1542	2.8948	1.1334	True
seamless_m4t+w2vbert+wav2vec_sb+cohere	4	cohere hubert	cohere	hubert	5.5591	7.8117	11.352	9.8465	2.2526	1.5055	True
whisper_v3+w2vbert+wav2vec_sb+cohere	4	cohere hubert	cohere	hubert	5.5591	7.8117	13.1172	11.0274	2.2526	2.0897	True
fastconformer+w2vbert+wav2vec_sb+cohere	4	cohere hubert	cohere	hubert	5.5591	7.8081	10.616	9.2461	2.249	1.3699	True
seamless_m4t+w2vbert+hubert+cohere	4	cohere fastconformer	cohere	fastconformer	5.5591	7.1898	7.3116	6.8969	1.6307	0.4147	True
seamless_m4t+w2vbert+wav2vec_sb+cohere	4	cohere fastconformer	cohere	fastconformer	5.5591	7.1898	11.352	7.8256	1.6307	3.5264	True
seamless_m4t+hubert+wav2vec_sb+cohere	4	cohere fastconformer	cohere	fastconformer	5.5591	7.1898	7.8379	6.8546	1.6307	0.9833	True

pool	k	pair	true_winner	pred_winner	wer_true_a	wer_true_b	wer_pseudo_a	wer_pseudo_b	gap_wer_true	gap_wer_pseudo	human_resolvable
whisper_v3+w2vbert+hubert+cohere	4	cohere fastconformer	cohere	fastconformer	5.5591	7.1898	9.0375	7.2857	1.6307	1.7519	True
whisper_v3+w2vbert+wav2vec_sb+cohere	4	cohere fastconformer	cohere	fastconformer	5.5591	7.1898	13.1172	8.9564	1.6307	4.1608	True
whisper_v3+hubert+wav2vec_sb+cohere	4	cohere fastconformer	cohere	fastconformer	5.5591	7.1898	9.335	7.1912	1.6307	2.1438	True
w2vbert+hubert+wav2vec_sb+cohere	4	cohere fastconformer	cohere	fastconformer	5.5591	7.1898	11.3123	7.6925	1.6307	3.6199	True
seamless_m4t+whisper_v3+w2vbert+wav2vec_sb+cohere	5	cohere fastconformer	cohere	fastconformer	5.5591	7.1898	8.709	7.7893	1.6307	0.9197	True
seamless_m4t+w2vbert+hubert+wav2vec_sb+cohere	5	cohere fastconformer	cohere	fastconformer	5.5591	7.1898	7.3633	6.9034	1.6307	0.46	True
whisper_v3+w2vbert+hubert+wav2vec_sb+cohere	5	cohere fastconformer	cohere	fastconformer	5.5591	7.1898	8.7079	7.3288	1.6307	1.3791	True
seamless_m4t+whisper_v3+w2vbert+cohere	4	cohere fastconformer	cohere	fastconformer	5.6229	7.2526	9.298	7.9155	1.6296	1.3825	True
seamless_m4t+whisper_v3+wav2vec_sb+cohere	4	cohere fastconformer	cohere	fastconformer	5.6229	7.2526	9.2367	7.8062	1.6296	1.4305	True
seamless_m4t+fastconformer+w2vbert+wav2vec_sb	4	fastconformer hubert	fastconformer	hubert	7.1898	7.8117	12.4435	10.1497	0.6219	2.2938	True
whisper_v3+fastconformer+w2vbert+wav2vec_sb	4	fastconformer hubert	fastconformer	hubert	7.1898	7.8117	13.9938	11.2815	0.6219	2.7123	True
seamless_m4t+whisper_v3+fastconformer+w2vbert+wav2vec_sb	5	fastconformer hubert	fastconformer	hubert	7.1898	7.8117	10.1231	10.0754	0.6219	0.0478	True
fastconformer+w2vbert+wav2vec_sb+cohere	4	fastconformer hubert	fastconformer	hubert	7.1898	7.8081	11.1109	9.2461	0.6182	1.8648	True
seamless_m4t+whisper_v3+fastconformer+w2vbert	4	fastconformer hubert	fastconformer	hubert	7.2526	7.8117	10.708	10.2688	0.5591	0.4393	True
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb	4	fastconformer hubert	fastconformer	hubert	7.2526	7.8117	10.5613	10.1819	0.5591	0.3794	True
whisper_v3+fastconformer+w2vbert+cohere	4	fastconformer hubert	fastconformer	hubert	7.2526	7.8081	9.473	9.3543	0.5555	0.1186	True

step7_floor_validated_ensemble_cv

model	frac_zero_wer_samples	kendall_b_all	kendall_b_nonzero	n_nonzero	kendall_b_editdist
cohere	0.8011	0.5541	0.3249	2088	0.5711
hubert	0.7598	0.674	0.592	2521	0.7008
fastconformer	0.757	0.6757	0.4825	2551	0.6876
seamless_m4t	0.6598	0.7692	0.595	3570	0.7861
whisper_v3	0.5688	0.8449	0.7169	4525	0.855
whisper_v2	0.4936	0.8641	0.7574	5315	0.8735
xlsr	0.4227	0.8876	0.8045	6059	0.9066
w2vbert	0.3718	0.8892	0.8214	6593	0.907
wav2vec_sb	0.317	0.9205	0.8736	7168	0.9367

step8_permutation_validated_ensemble_cv

perm	order	n	identical	differing	identical_frac
0	cohere+wav2vec_sb+hubert+w2vbert+fastconformer+whisper_v3+seamless_m4t	10495	10495	0	1
1	fastconformer+hubert+w2vbert+seamless_m4t+whisper_v3+cohere+wav2vec_sb	10495	10495	0	1
2	hubert+cohere+w2vbert+wav2vec_sb+seamless_m4t+fastconformer+whisper_v3	10495	10495	0	1
3	cohere+seamless_m4t+hubert+fastconformer+w2vbert+whisper_v3+wav2vec_sb	10495	10495	0	1
4	wav2vec_sb+w2vbert+fastconformer+seamless_m4t+whisper_v3+cohere+hubert	10495	10495	0	1
5	cohere+wav2vec_sb+seamless_m4t+w2vbert+fastconformer+whisper_v3+hubert	10495	10495	0	1